KavachX AI/ML Project Report

IMPLEMENTATION AND MEASURED DEMONSTRATIONS | 26 OCTOBER 2026

Three implemented workflows

Task 1 - English rules to event JSON. Rule parsing, YOLO-World open-vocabulary detection, ByteTrack identities, polygon dwell timing, privacy-masked evidence and schema-validated events. Demonstrated bag and car rules using the same model weights, plus a short-video control.

Task 2 - PPE training and quantization. YOLOv8n trained at 416 pixels for 15 epochs on 240 training and 60 validation images. Includes a trained checkpoint, training logs/plots, five reviewed examples, FP32/FP16/INT8 exports and static PTQ using 100 calibration images.

Task 3 - Secondary snapshot verification. Local CLIP ViT-B/32 with prompt ensembles, a calibration-selected threshold, 30-image test evaluation and per-image scores/decisions. Includes monthly-cost and reviewed-feedback workflow notes.

PPE model measurements

Model	MiB	CPU ms	mAP@50	Helmet R	No-helmet R
FP32	11.580	12.390	0.501414	0.656595	0.294118
FP16	5.829	13.419	0.501651	0.655862	0.294118
INT8	3.197	7.411	0.491185	0.660159	0.235294

Apple M4 CPU; ONNX Runtime; batch 1; four threads; 10 warmups and 100 timed runs. Latency is model execution. INT8 is approximately **3.62x smaller** and **1.67x faster** than FP32 on this host, with a 1.023-point mAP difference.

Snapshot verifier measurements

Pipeline	TP	FP	FN	TN	Precision	Recall
Forward all	15	15	0	0	50.0%	100.0%
CLIP verifier	11	6	4	9	64.7%	73.3%

Threshold: **0.3187942803**, selected using 20 calibration images and evaluated on 30 separate test images. Warm CPU scoring: **18.79 ms average**, 19.48 ms p95, including image decoding/preprocessing/scoring.

Execution and handover

Task 1 demonstration evidence

One bag candidate was emitted at video time **34.3 seconds** after **10.1 seconds** of qualifying dwell, with **needs_verification=true**. The car control emitted one event at 3.1 seconds; the five-second control emitted zero. Independent identity agreement covered 143 observations, with mean box IoU 0.8557.

Run the project

Use Python 3.12 and follow the included README.md. The checkpoint and ONNX exports are included. First use downloads CLIP weights. Task 2 training has a separate GPU workflow; the demonstration commands use the existing checkpoint.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m unittest discover -s tests -v
```

The README provides the dataset-root update and exact execution commands for all three tasks. KavachX_Run_All_Tasks.ipynb supplies the Colab launcher and creates a separate Python 3.12 task environment. Its upload cell accepts this same project ZIP.

Verification and included artifacts

Recorded checks: **29 passing regression tests** (17 Task 1, six Task 2, six Task 3). Fresh Task 3 inference reproduced all 50 saved scores exactly in a separate local environment. Source hashes, annotation-derived labels, calibration/test recording groups and packaged artifacts were audited.

The ZIP includes source modules, tests, pinned direct dependencies, trained PPE weights, ONNX variants, demonstration assets, manifests, attribution, training logs, plots, event examples, masked evidence, score tables and run instructions. The reported measurements are saved local CPU results.

Operating model and source attribution

At 200 alerts/day for 30 days, local CLIP performs **6,000 monthly verifications** with **$0 hosted inference API charges**. Hardware and operations are accounted for separately. Reviewed dashboard feedback can be stored with event/model versions and used in offline evaluation.

Demonstration sources: **MOTChallenge MOT17-04**, **Ultralytics Construction-PPE**, and **BIRDSAI real nighttime thermal images**. These are attributed substitute assets; BIRDSAI labels indicate human presence. Source and dataset terms are retained in the task folders. Share the package privately.

AI assistance: OpenAI Codex assisted implementation, source discovery, debugging, verification and documentation. The candidate should understand and explain the code. The package is an implementation handover; this report does not make a blanket claim that every assessment requirement has been completed.